

Computational Supplement to “Future-Horizon Regularity and One-Sided Radiative Traces Do Not Select the Einstein Sector of Pure Weyl Gravity”

GPT-5.6.sol* Asger Alstrup Palm[†]

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1 Purpose

This supplement is the destination for verification details removed from the main narrative: recurrence bounds, interval cells, projective chart changes, contour panels, mutation tests, content hashes, and exact commands. The theorem statements and proofs in the main paper are the mathematical authority. The machine-readable certificates provide pinned reproducibility records, exact witnesses, validation margins, and independent implementation checks. A timeout, skipped verification, or reproduced producer output is never recorded as an independent verification pass.

2 Logical dependency of the title theorem

The title theorem is computer-assisted but has a closed proof chain. It does not use an explicit T_+ matrix, a retarded convolution, the separately certified finite-interval radial Fredholm pole, or a raw-Weyl/current comparison not already cited as an endpoint authority.

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Result	Proof dependency	Status and domain	Title theorem
Gauge-invariant carrier discriminator	Schouten carrier modulo source diffeomorphism/Weyl gauge; nonzero Maxwell two-form witness	Exact; Ricci-flat background	Required
Axial carrier filtration	RW/RW/Maxwell exact filtered system and quotient maps	Exact; Schwarzschild, axial $\ell = 2$	Required
Incoming realization	Scalar Wronskians, boundary dévissage, $T_- \in GL(3, \mathbb{C})$	Analytic plus certified computation; every real $\omega > 0$	Required
Outgoing realization	Triangular determinant and nonzero scalar outgoing factors	Validated; [1/2, 10001/20000]	Required only for outgoing clause
Finite-interval radial QNM Green pole	Analytic initial-value reduction and certified Smith transfer	Exact reduced-mode theorem; no causal inverse transform	Not required
Explicit T_+ , simultaneous pure-outgoing non-Einstein mode, and retarded/causal extensions	Separate programme items	Not established	Not required

3 Evidence taxonomy

Evidence kind	Required supplement record
Exact symbolic	input hashes, exact arithmetic kernel, canonical witness, independent verifier
Validated numerical	declared domain, interval backend, nonzero margins, subdivision policy, independent verifier
Numerical fixture	precision, residuals, purpose, and explicit non-theorem status
Open computation	last valid checkpoint, failure mode, and <code>does_not_establish</code> ledger

3.1 Independence profile

Independence is recorded as four separate Boolean dimensions, not as a hierarchy:

Dimension	Meaning
Independent code	Separate executable rather than a producer rerun
Independent representation	Different elimination, variables, or mathematical representation
Independent backend	Different exact-arithmetic, interval, or CAS backend
Independent derivation	Independently derived mathematical formulation

Rerunning a producer is reproduction, not verification. Each manifest records its independence profile and all shared dependencies.

Every publication record pins

(repository commit, certificate SHA-256, producer SHA-256, verifier SHA-256)

together with dependency versions, arithmetic or interval backend, platform metadata where relevant, the canonical command, expected exit status, and deterministic generator definition.

4 Trusted computing base

Exact symbolic certificates rely on Python integer/rational arithmetic or on the exact backend named in their manifests; every denominator assumption and parameter domain is part of the corresponding certificate. Validated numerical certificates use outward-rounded interval endpoints, retain the complete enclosing interval in machine-readable form, and state the subdivision and refusal policy. JSON parsing, the Python runtime, the declared exact/interval kernel, and the operating system are inside the trusted computing base unless an independent-backend check explicitly removes one of them.

The publication release must contain an immutable archive identifier in addition to the Git commit and file hashes. No DOI has yet been assigned, so this working draft does not claim that archival gate.

5 Concrete title-theorem records

Claim	Primary record and SHA-256	Domain/margin	Independence profile
Carrier discriminator ff86f50e88d07c0ec8 fcd8ca4788e60058db 07f90a7a96a9be5164 e033952908	phase4/covariant_einstein_maxwell_carrier_v1/certificate.json Exact Ricci-flat identities; declared gauge quotient	code, representation, derivation	
Axial filtration 1a6ac48dcc52659997 cb0ed47709117308ef 258c05e6996c559d9a d81f9ae579	phase3/axial_rw_lx_triangular_preflight/certificate.json Exact, $M = 1, \ell = 2$; denominators listed in record	code, representation	
Incoming realization f223358ca9de0f6d81 9684ce61d62677d6e5 f8c5d4edaa600e2bae 02719af0ef	phase3/axial_incoming_extended_domain_audit/certificate.json Every real $\omega > 0$; Wronskian/déviage margins in record	code, representation	
Outgoing cell de7e3a944e945b00fe a9e2b5f7abffc8789c 7449cae9eae222c047 912a4f3d0f	phase3/axial_outgoing_population_cell_half_v1/certificate.json $ A_{\text{out},2} > 0.0113262, A_{\text{out},1} > 0.146805$	code, representation, interval verifier	
Radial QNM Green pole e3f5e28d426305f0e6 8f7c11c7f63e1f0618 cc63ad20fc1ff5b221 fb38d34110	phase4/axial_qnm_fredholm_promotion_v1/certificate.json One certified QNM disk; rank-one order-two finite-interval radial inverse	code, representation	

The paths above are relative to `black_hole_programme/`. The machine-readable records contain the complete intervals, producer/verifier commands, expected exit status, dependencies, and mutation results; the decimals in the table are readability-only lower bounds.

6 Principal certificate chains

The publication dependency set consists of the following independently audited chains:

1. the established conformal detour context, the universal Ricci-factorized Hessian, the Schouten–Einstein/Maxwell carrier, and its Euler-current boundary;
2. the second-order auxiliary-tensor parent action, its factorized Einstein Green current, canonical null lift, and QNM divisor count;
3. the parent-Hessian inverse, rank-one QNM Laurent algebra, simple-self-extension involution lemma, and endpoint Krein graph/cotangent obstructions;
4. axial operator extraction and Ricci reconstruction;
5. RW/RW/spin-one factorization, partial-jet identities, and the exact no-rational-intertwiner test;
6. incoming and horizon Lee–Wald Grams;
7. global incoming population and the common scattering cell;
8. boundary dévissage and the no-growing-mode theorem;
9. projective Evans winding, local selector, spin-one local-unit test, and finite-interval radial Fredholm promotion.

The primary local-structure reproducibility record is `black_hole_programme/phase4/axial_universal_hessian_intertwiner_v1/certificate.json`. Its independent verifier re-derives the curvature-invariant coefficients, the 2×2 positivity determinant, the fourth-order intertwiner compatibility equation in both spin directions, the indicial factors, and the terminal $A + B/r$ substitution. Five mutation tests fail closed on the Hessian trace coefficient, horizon exponents, terminal residual, and a forbidden nonlocal promotion.

The parent-current reproducibility record is `black_hole_programme/phase4/second_order_parent_flux_v1/certificate.json`. Its independent-representation verifier independently repeats the four-dimensional trace elimination and null-lift algebra, checks the imported Euler and physical connection-EP2 boundaries, and fails closed on generic radial-to-time-Jordan promotion, Green-resolvent promotion, all-frequency reflection-zero exclusion, and a complete polar Gram. The principal future current audit is

$$G_{\text{raw Weyl}} - G_{\text{parent}} = G_{\text{Euler cut}}$$

at the horizon, a finite matching section, and null infinity. Until those endpoint comparisons are separately certified, the universal parent identity does not substitute for the existing raw-Weyl endpoint certificates.

6.1 Parent-resolvent and Krein-obstruction chain

The exact record `black_hole_programme/phase4/parent_resolvent_krein_obstructions_v1/certificate.json` checks the auxiliary-tensor Hessian inverse by two-sided block multiplication and independently verifies the rank-one identity

$$PAP = \frac{\langle \tilde{u}, \mathcal{A}u \rangle}{\alpha^2} u \otimes \tilde{u}.$$

It also certifies the characteristic-zero involution lemma for a nonsplit self-extension of a simple object, the endpoint positive-graph obstruction, and the quotient duality $B_2/E \simeq E^*$.

That algebraic record is deliberately fail-closed at a differential QNM pole. The separate `black_hole_programme/phase4/axial_qnm_fredholm_promotion_v1/certificate.json` closes the fixed-finite-interval radial gate by exact variation of constants, boundary Schur reduction, and Smith transfer. It certifies a nonzero rank-one second-order radial inverse pole and survival under exact metric reconstruction. It remains fail-closed at the causal exterior spacetime resolvent, inverse-Laplace deformation, and generalized ringdown statement. Real-axis simplicity and scalar endomorphism rings are independently certified in `black_hole_programme/phase4/rw_maxwell_simplicity_endomorphisms_v1/certificate.json`. Consequently the statement that every rational involution of the nonsplit axial spin-two block is $\pm I$ is unconditional for real $\omega > 0$, while the complex-frequency classification remains open.

7 Computations excluded from the publication proof

The explicit- T_+ , parent-overlap, retarded-convolution, and Schwarzschild time-domain calculations are maintained in the programme reports rather than in this paper-specific supplement. They are not dependencies of the title theorem.

The most recent explicit- T_+ attempt has the concrete fail-closed record `black_hole_programme/phase4/axial_explicit_tplus_band_v1/amplitude_certificate.json`. It preserves the physical endpoint-amplitude cocycle and reaches $r = 4$, but its rectangular terminal remainders exceed the coefficient scale by more than 10^6 in both frames. It therefore certifies neither the terminal inverse nor T_+ . This is a shortfall of that enclosure representation, not evidence that T_+ is singular. Its certificate SHA-256 is `d8322b37a637e12cd078872d22c083c0b109ede0dcfc71b3ba6b230551aafd43`.

8 Explicit non-results

Neither the main paper nor this supplement establishes:

- a non-Einstein separated solution that is simultaneously future-horizon regular and purely outgoing at infinity;
- explicit T_+ entries or reflection amplitudes;
- a causal exterior spacetime resolvent, inverse-Laplace deformation, generalized ringdown term, or retarded convolution on Schwarzschild;
- a complete polar global connection;
- a canonical positive quantum metric or unitarity.